Integrated Lecture · XAI

Explainable Machine Learning

From SHAP, LIME and Counterfactuals to Financial Regulation

Explainable Machine Learning for Finance

Merges the XAI content of ten past lecture materials without duplication, updated with research and regulation as of October 2026.
Hand-calculation examples are collected in a separate workbook deck (XAI_Workbook).

Explainability has three faces: decomposition, approximation, obligation

We state the conclusion that runs through the whole lecture first

1

Explanation is attribution

SHAP splits a single prediction exactly into a base value plus per-feature contributions.

2

Explanations are approximations and can disagree

Post hoc explanations approximate the model. Methods reach different conclusions and can even be deliberately fooled.

3

Explanation is now a regulatory duty

The EU AI Act, GDPR case law, Korea's AI Basic Act and the Financial AI Guidelines require individual explanations.

Key question: how do we tell customers, supervisors and developers why the model decided as it did?

The lecture moves from definitions to tools, cases and regulation

We proceed in eight parts

Part	Topic	Key tools and questions
1	Why explainability	Definitions, taxonomy, four levels of explanation, three audiences
2	Global explanation	Coefficients, tree & permutation importance, surrogates, PDP·ALE
3	Shapley values and SHAP	Axioms, additivity, background distribution, estimators, interactions
4	Reading SHAP plots	waterfall, beeswarm, scatter, decision plot
5	Local surrogates · rules · counterfactuals	LIME, Anchors, counterfactual explanations
6	Deep learning and LLMs	Integrated gradients, attention weights, circuit tracing
7	Financial applications	Credit scoring, fraud detection, asset price prediction
8	Limits and regulation	Disagreement, explanation manipulation, 2026 regulation

01

PART 1 · Basic Concepts

Why explainability

Definitions, taxonomy, depth of explanation, and the three audiences of an explanation

Interpretability and explainability are different concepts

We separate interpretability from explainability

Interpretability is how far a person can read the model's structure itself, explainability is the ability to give a person the grounds for an individual decision.

Interpretability

- Linear regression coefficients, shallow decision trees
- The decision rule is visible from the structure
- The price is limited expressive power

Explainability

- Post hoc explanations attached to black boxes such as boosting and neural nets
- SHAP, LIME and counterfactual explanations are typical
- Never forget that an explanation approximates the model

XAI methods are classified along four axes

The same SHAP can serve as a local or a global explanation depending on how it is used

Axis	Categories	Example
Scope	Global vs local	Overall importance ranking vs one customer's rejection reasons
Timing	Intrinsic vs post hoc	Logistic coefficients vs SHAP
Target	Model-specific vs model-agnostic	TreeSHAP vs KernelSHAP·LIME
Form	Attribution, rule, counterfactual, example	SHAP, Anchors, DiCE, prototypes

This lecture widens the form of explanation in the order ‘attribution → rule → counterfactual’

Explanation deepens in four levels: result, ingredients, contribution, change

Cooking-show analogy: how far does an AI chef explain why it made the cake?

1. Result only

"Delicious cake, done"

"Your loan was declined."
No reason, so trust is low.

Low trust

2. Ingredient list

"Flour, sugar, eggs"

"We considered income and credit score."
What mattered most is unknown.

Moderate trust

3. Contribution

"Flour 30%, sugar 25%"

"Income contributed most; delinquencies lowered the score."

SHAP·LIME

4. Change simulation

"What if less sugar?"

"Repaying existing debt would clear the approval threshold."

Counterfactual explanation

Customers ask both ‘why’ and ‘what should I do’

In finance, explanations serve regulation, trust and model improvement

Source: SHAP guide in an earlier lecture

Regulatory compliance

The grounds of automated decisions must be shown to customers and supervisors. Core evidence in model risk management documentation.

Details in PART 8

Business trust

To an executive asking "Why did we block this customer?", we can answer "credit score, late-night and large transactions."

Fraud detection practice

Model improvement

Remove features that contribute almost nothing, and find biased features and data quality issues.

Details in PART 7

The same decision needs different explanations for different audiences

When designing an explanation, first decide who reads it

Audience	What they want to know	Suitable explanation
Customer	Why declined, what to change	Top 2–4 reasons, counterfactual guidance
Supervisor	Procedures and principles, fairness, validation records	Documentation, global importance, validation report
Developer · validator	Errors, data leakage, bias	Dependence plots, false positive/negative pattern analysis

Hand the same SHAP chart to all three audiences and nobody is satisfied

02

PART 2 · Global Explanation

Intrinsic interpretation and permutation importance

First see which features the model as a whole relies on

In linear models, feature importance is read from coefficient size

Source: Big Data Ch. 5 machine learning slides. β_j is the coefficient of feature j, σ(x_j) its standard deviation

$$\text{Importance}_j = \underbrace{|\beta_j|}_{\text{if standardized}} \qquad \text{Importance}_j = \underbrace{\sigma(x_j)\,|\beta_j|}_{\text{if not standardized}}$$

How to read

- With standardized features, the absolute coefficient is the importance
- Otherwise multiply by the standard deviation to align units

Pitfall

- High multicollinearity makes coefficients unstable
- Worst in OLS; ridge and lasso suffer less

Default tree importances can be biased toward training data

Three metrics used in regression trees, random forests and boosting

Metric	Definition	Caution
Split count	Number of times the feature is used as a split variable	Overrates continuous variables with many distinct values
Impurity decrease (MDI)	Average reduction in prediction error at splits	Computed on training data, so it reflects overfitting
OOB permutation (random forest)	Increase in OOB error after shuffling the feature	Correlated features share importance

Generalizing OOB permutation gives the model-agnostic permutation feature importance on the next slide

Permutation feature importance is the loss increase when a feature is shuffled

L is the loss function (squared error, 1−R², etc.); x_perm,j is the data with only feature j randomly shuffled

$$FI_j = \underbrace{L\left(y, \hat{f}(x_{\text{perm},j})\right)}_{\text{loss after shuffling feature } j} - \underbrace{L\left(y, \hat{f}(x)\right)}_{\text{original loss}}$$

Procedure

- Measure loss on the original data
- Shuffle only feature j and measure again
- The loss increase is the importance

Pros and limits

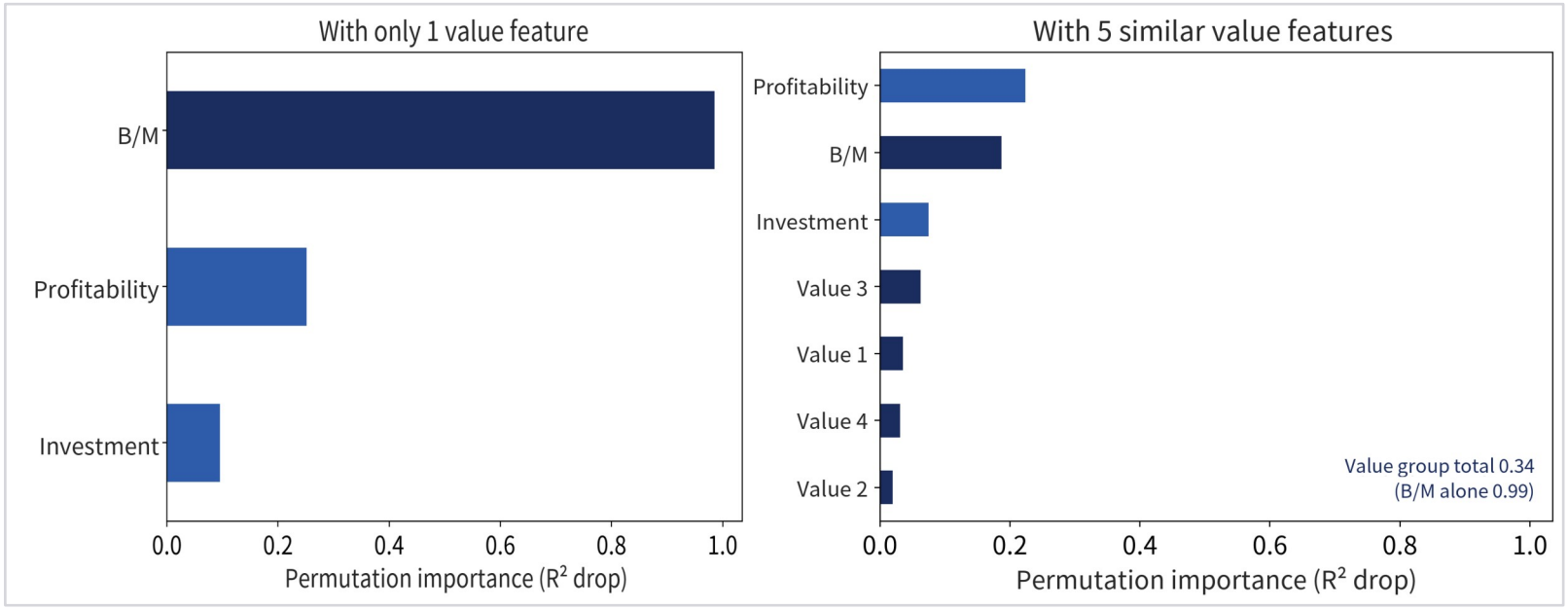
- Model-agnostic and simple to compute
- Highly correlated features are underrated
- Gives global importance only

Adding several similar features splits importance and makes each look small

Synthetic experiment: return = B/M + 0.5×profitability + 0.3×investment + noise, random forest

Substitution effect

- Reproduces the source's warning
- B/M alone 0.99
- With 5 value vars, B/M 0.19
- Group total 0.34
- Aggregate correlated features as a group



A global surrogate re-approximates the black box's predictions

Global surrogate: trained on black-box predictions instead of the actual target

$$\underbrace{\hat{f}(x_i)}_{\text{black-box prediction}} = \underbrace{\hat{f}_{\text{surrogate}}(x_i)}_{\text{interpretable surrogate model}} + \varepsilon_i$$

Procedure

- Collect black-box predictions
- Fit OLS or a shallow tree to those predictions
- Check surrogate fidelity with R^2

Caution

- A surrogate's explanation is not the original model's explanation
- With low fidelity, interpretation is meaningless

PDP shows average effects, ALE shows local effects

S is the feature of interest, C the other features, z_k the interval edges, N_k the samples in interval k

$$\underbrace{\hat{f}_S(x_S)}_{\text{partial dependence function}} = \frac{1}{n}\sum_{i=1}^{n} \hat{f}\left(x_S,\ x_C^{(i)}\right)$$

$$\hat{f}_{\mathrm{ALE}}(x) = \sum_{k=1}^{k(x)} \underbrace{\frac{1}{n_k}\sum_{i\in N_k}\left[\hat{f}\left(z_k, x_C^{(i)}\right) - \hat{f}\left(z_{k-1}, x_C^{(i)}\right)\right]}_{\text{average local change within interval } k}$$

Partial dependence (PDP) · ICE

- Keep other features at data values, vary only the feature of interest, and average
- With strong correlation, it creates unrealistic combinations

Accumulated local effects (ALE)

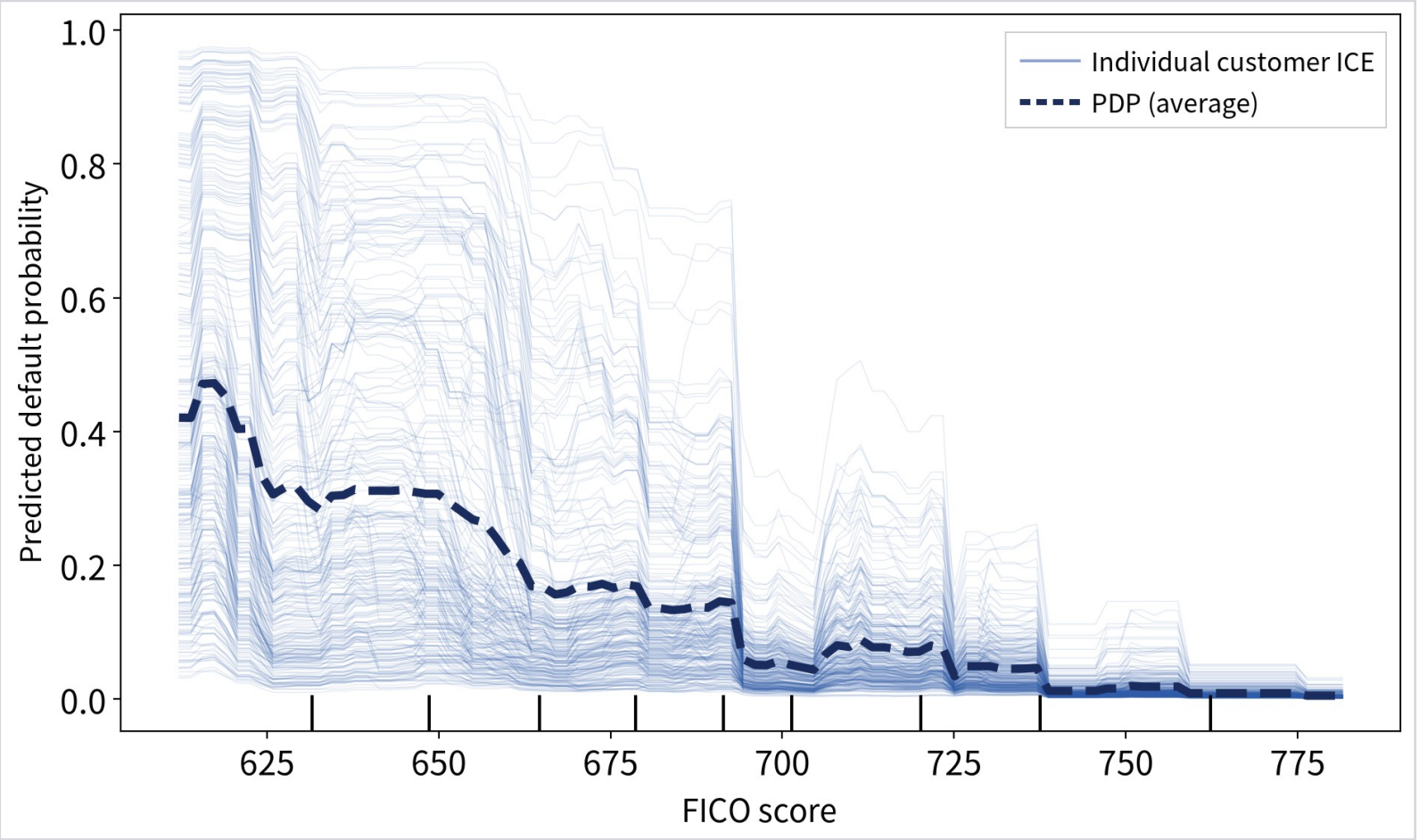
- Measures change only within intervals where data exist
- Robust to correlated features (Apley·Zhu, 2020)

ICE lines reveal customer-level differences hidden behind the average PDP

Synthetic loan data, LightGBM default model (AUC 0.87), PDP and ICE for FICO score

How to read

- Thick dashed line: average effect
- Each thin line: one customer
- Sharp drop near score 700
- Crossing lines mean interaction



03

PART 3 · Shapley Values

Shapley values and SHAP

How fair division in cooperative games became prediction explanation

Shapley value: game theory's fair split of cooperative gains

Episode and intuition first, then the formula

Episode

- Lloyd Shapley proposed it in 1953 to solve payoff division in cooperative games
- He received the 2012 Nobel Prize in Economics
- Some 60 years later, in 2017, Lundberg·Lee revived it as the standard for ML explanation

Pizza order problem (source example)

- Ordered separately, three people pay 10,000 won each (30,000); together, 25,000 won
- How should the 5,000 won discount be returned to each person?
- Averaging each person's added share over all joining orders is fair

Replace players with features and the Shapley value becomes SHAP

Apply the same frame to decomposing model performance (R^2) and you get global importance

Game theory	Local explanation (SHAP)	Global explanation (R^2 decomposition)
Player	Feature	Feature
Value of coalition S, v(S)	Expected prediction knowing only S	R^2 of the model using only S
Payoff φ_i	SHAP value of feature i	R^2 contribution of feature i
Total value	Prediction – base value	R^2 of the full model

Source Big Data Ch. 5 covered the global use with R^2 as value; earlier lectures covered the local use

The Shapley value weights marginal contributions over all orders

N is the full feature set, n the number of features, S a subset excluding feature i

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \underbrace{\frac{|S|!\,(n - |S| - 1)!}{n!}}_{\text{ordering weight}} \underbrace{\Big[v(S \cup \{i\}) - v(S)\Big]}_{\text{marginal contribution of feature } i}$$

Intuition

- Features join one by one in random order
- Average i's added share over all orders

Computational burden

- With 2^n subsets, exact computation soon becomes infeasible
- Hence the estimators later in PART 3

R^2-decomposition Shapley values use the same formula

Source Big Data Ch. 5 notation: K features, C a coalition, V(C) the R^2 of the model using coalition C

$$SV_j = \frac{1}{K} \sum_{C \subseteq \tilde{K} \setminus \{j\}} \binom{K-1}{|C|}^{-1} \Big(\underbrace{V(C \cup \{j\}) - V(C)}_{R^2 \text{ gain from adding } j \text{ to the coalition}} \Big)$$

How to read

- Average within each coalition size, then average again over the K sizes
- Equal to the ordering weights on the previous slide

Practical implications

- Exact computation is feasible for OLS with few features
- With many features, approximation is needed

The Shapley value is the unique allocation satisfying four axioms at once

Efficiency underlies SHAP's additivity, the property that ‘it adds up’

$$\sum_{i=1}^{n} \phi_i = v(N) - v(\varnothing)$$

Axiom	Meaning
Efficiency	Contributions sum to the difference between total value and base value
Symmetry	Two features adding the same amount to every coalition receive the same value
Dummy	A feature that contributes to no coalition receives 0
Additivity	Adding two games adds their Shapley values

SHAP splits a prediction into a base value plus contributions

The base value φ_0 is the average prediction over background data

$$\hat{f}(x) = \underbrace{\phi_0}_{\text{base value} = E[\hat{f}(X)]} + \sum_{i=1}^{M} \underbrace{\phi_i}_{\text{SHAP value of feature } i}$$

Source fraud detection example (corrected): starting from a 10% base value, the four feature contributions add up to 85%

$$\underbrace{0.10}_{\text{base}} + \underbrace{0.10}_{\text{age}} + \underbrace{0.05}_{\text{income}} + \underbrace{0.30}_{\text{credit score}} + \underbrace{0.30}_{\text{transaction time}} = \underbrace{0.85}_{\text{fraud probability}}$$

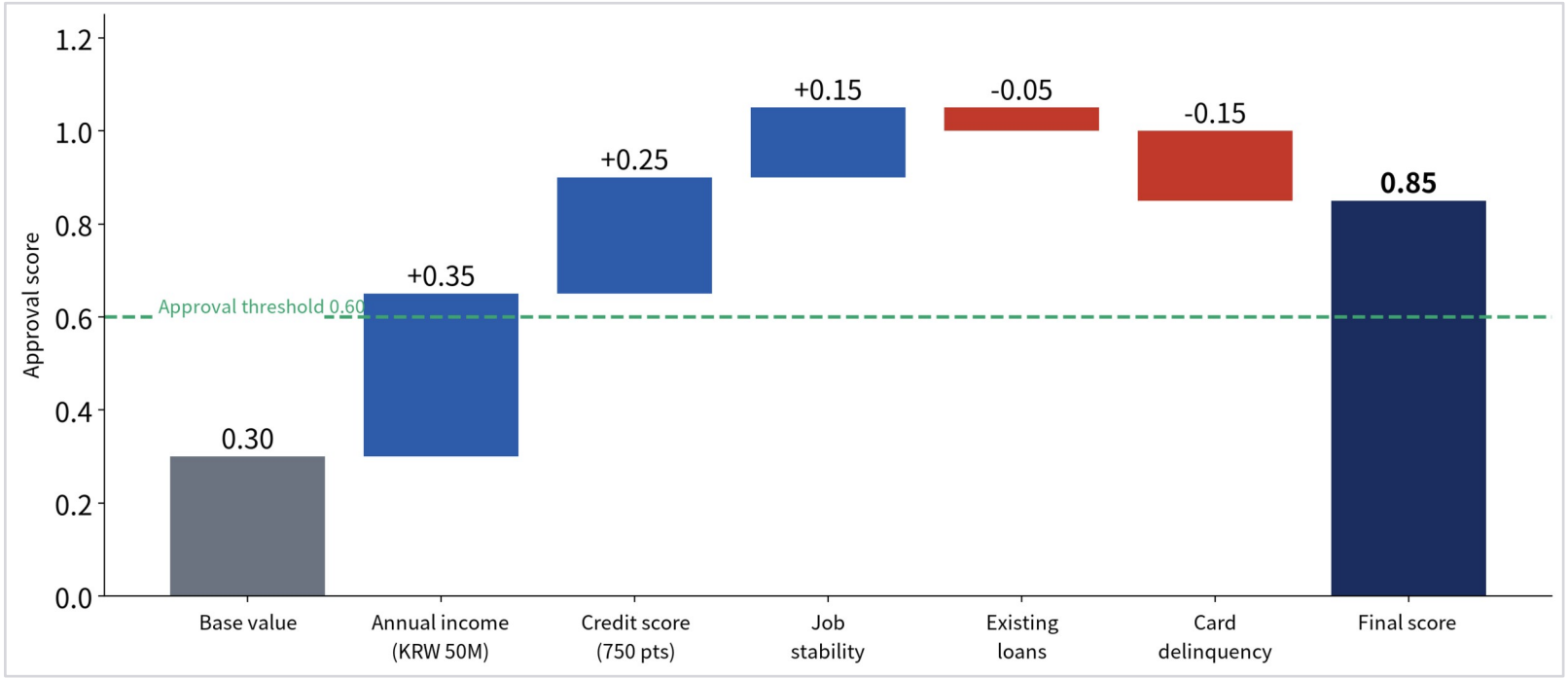
The source listed the transaction-time contribution as +0.40, summing to 0.95. Corrected to +0.30 to match 85%

A hypothetical customer's approval score builds from base 0.30 to 0.85

Hypothetical case Mr. Kim Cheol-su from an earlier lecture. Total 0.30+0.35+0.25+0.15−0.05−0.15=0.85

What to tell the customer

- Top approval reason: annual income
- Top deduction: card delinquency
- Repaying debt adds headroom
- Numbers are hypothetical



SHAP values depend on how missing features are filled in

The ‘true to the model or true to the data’ question (Chen et al., 2020; Janzing et al., 2020)

$$v_{\text{conditional}}(S) = E\left[\hat{f}(X) \,\middle|\, X_S = x_S\right]$$

$$v_{\text{interventional}}(S) = E_{X_C}\left[\hat{f}(x_S, X_C)\right]$$

Conditional (observational)

- Uses remaining values consistent with the known feature values
- Reflects data correlation, but features unused by the model also get credit

Interventional

- Draws remaining features independently from background data
- Faithful to how the model actually works
- TreeExplainer with background data uses this approach

Matching the estimator to the model buys both speed and accuracy

Main Explainers in the shap library. T is the number of trees, L leaves, D depth

Estimator	Target model	Accuracy and speed
TreeExplainer	Random forest, XGBoost, LightGBM, CatBoost	Exact, polynomial time $O(TLD^2)$
KernelExplainer	Any model (black box)	Weighted-regression approximation; slow and varies with sampling
DeepExplainer	Neural nets (TensorFlow, PyTorch)	Approximation combining DeepLIFT with Shapley values
LinearExplainer	Linear · logistic regression	Closed form, instant
Permutation·Partition	Any model (default Explainer)	Permutation approximation, groups correlated features

Linear SHAP is closed form; Kernel SHAP is weighted regression

M is the number of features, |z′| the number switched on. Weights are large for very small or very large coalitions

$$\phi_i = \beta_i \left(x_i - E[x_i] \right)$$

$$\pi(z') = \frac{M - 1}{\binom{M}{|z'|} \; |z'| \; (M - |z'|)}$$

Linear SHAP

- Multiply the coefficient by the deviation from the mean
- Values above the mean contribute in the coefficient's direction

Kernel SHAP

- Sample feature subsets and predict
- Solve a linear regression with these weights; the coefficients are the SHAP values

Recent shap versions changed the return format of SHAP values

List indexing like shap_values[1] in the source lecture code must be rewritten for new versions

Version	What changed
0.45	Multi-output SHAP values returned as an array (samples×features×classes) instead of a list
0.47	Legacy bar plot removed; reorganized around the Explainer API
0.48	Added CoalitionExplainer and Winter values
0.54.1	Released 16 Oct 2026; this deck's figures were drawn with this version

[Output] Recommended code · shap 0.54

```
explainer = shap.TreeExplainer(model)
sv = explainer(X_test)   # Explanation

sv.values.shape  # LightGBM binary:
                 # (n, p) log-odds
                 # sklearn RF: (n, p, 2)

shap.plots.waterfall(sv[0])
shap.plots.beeswarm(sv)
```

SHAP interaction values isolate pairwise effects

TreeSHAP interaction values of Lundberg et al. (2020). M is the number of features

$$\phi_{ij} = \sum_{S \subseteq N \setminus \{i,j\}} \frac{|S|!\,(M - |S| - 2)!}{2\,(M-1)!} \underbrace{\left[v(S \cup \{i,j\}) - v(S \cup \{i\}) - v(S \cup \{j\}) + v(S)\right]}_{\text{extra effect when both features are present}}$$

How to read

- Diagonal entries are main effects, off-diagonal entries pairwise interactions
- Main effects plus all interactions add up to the prediction

Latest trend: shapiq

- Handles higher-order interactions among three or more features
- Muschalik et al., NeurIPS 2024 library

04

PART 4 · Reading the Plots

Reading SHAP plots

We draw and read real plots with the same four features as the credit model in the source conversation

Each SHAP plot answers a different question

Local plots show one case; global plots show the overall pattern

Plot	Scope	Question answered
waterfall	Local	How was one prediction built up from the base value?
force	Local	What forces push the prediction up or down?
beeswarm (summary)	Global	Which features matter, and which way do high values push?
bar	Global	How do features rank by mean absolute SHAP value?
scatter (dependence)	Global	How do feature values relate to contributions; any interactions?
decision	Local, many cases	How do the cumulative paths of many cases diverge?

A LightGBM default model on synthetic four-feature loan data

Same feature set as the figures in the source conversation: DTI, FICO, income, home ownership

4,000	18%	0.87	−2.40
Synthetic loan samples (30% test)	Default rate	Test AUC	Base value (default log-odds, prob. about 8%)

Mind the units

- TreeExplainer's default output is log-odds, not probability

Additivity check

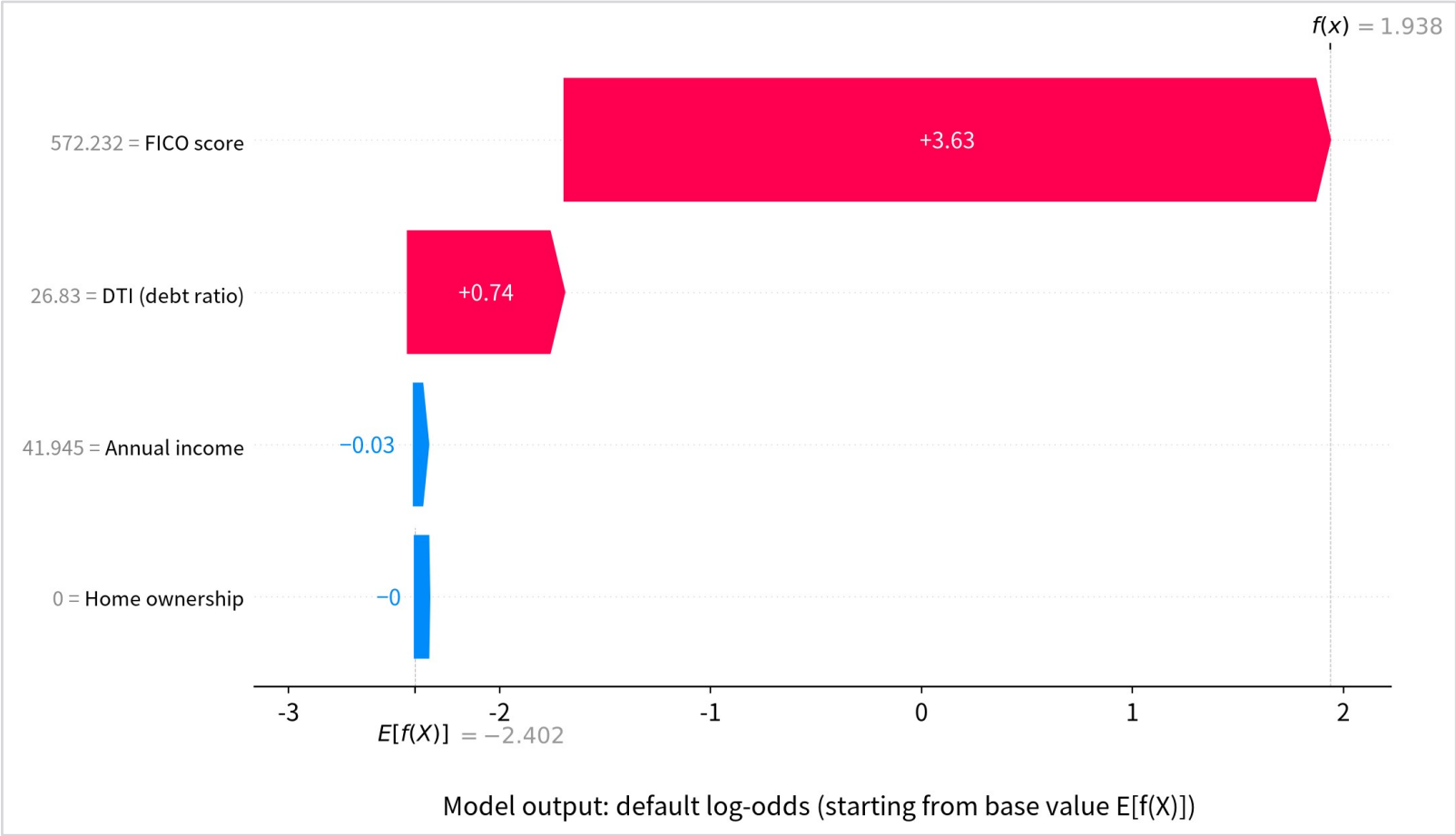
- Base value plus SHAP sum matched the predicted log-odds to 14 decimal places

A waterfall shows how one customer's prediction is built up

One high-risk customer: from base −2.40, FICO +3.63 and DTI +0.74 give 1.94 (default prob. about 87%)

How to read

- E[f(X)] at the bottom is the start
- Red raises risk
- Blue lowers risk
- f(x) at the top is the prediction

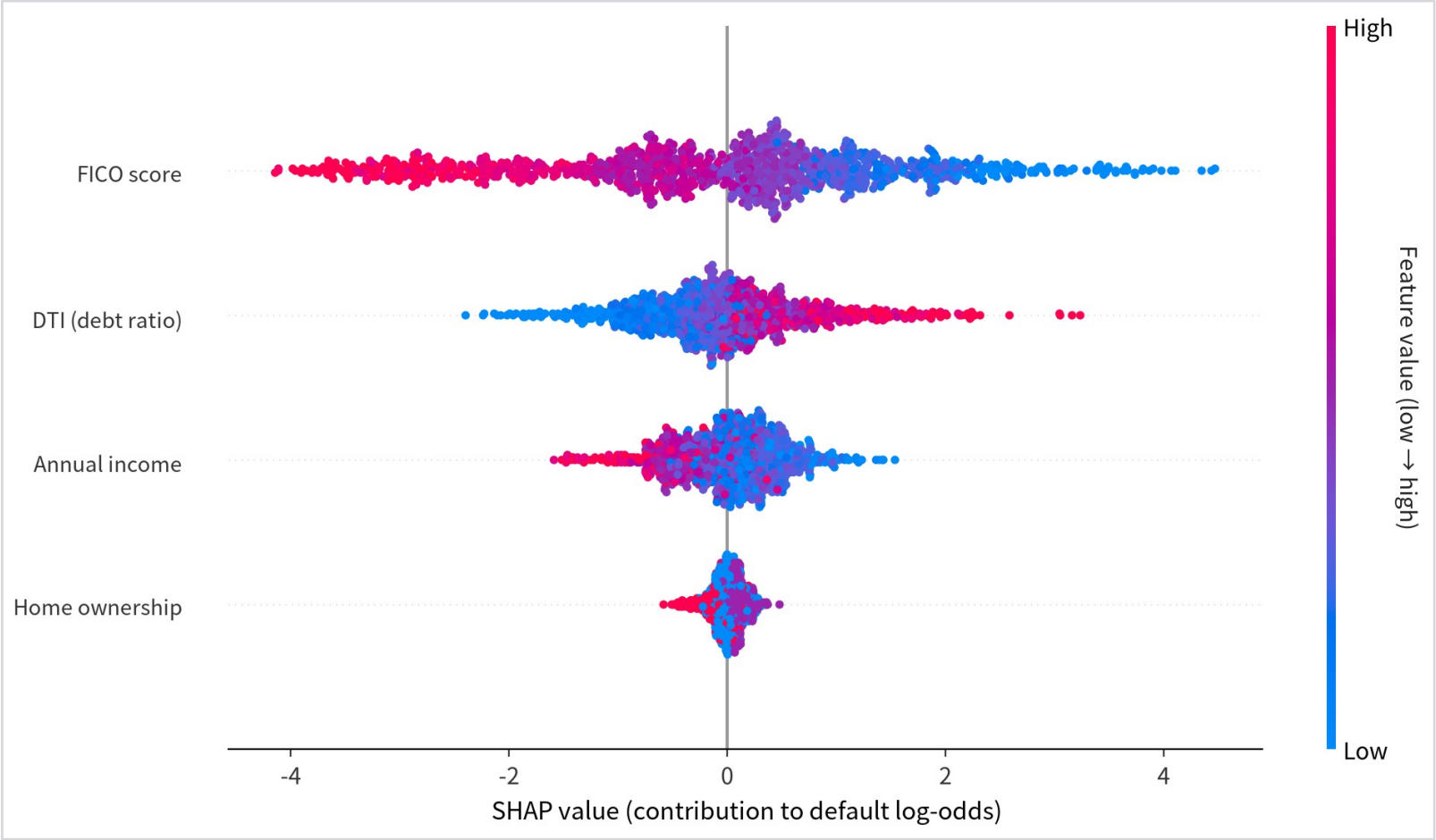


A beeswarm shows importance and direction in one chart

Each dot is one customer; color is the feature value, horizontal position the SHAP value

How to read

- Higher means more important
- Low FICO (blue) goes right
- High DTI (red) goes right
- Home ownership matters little

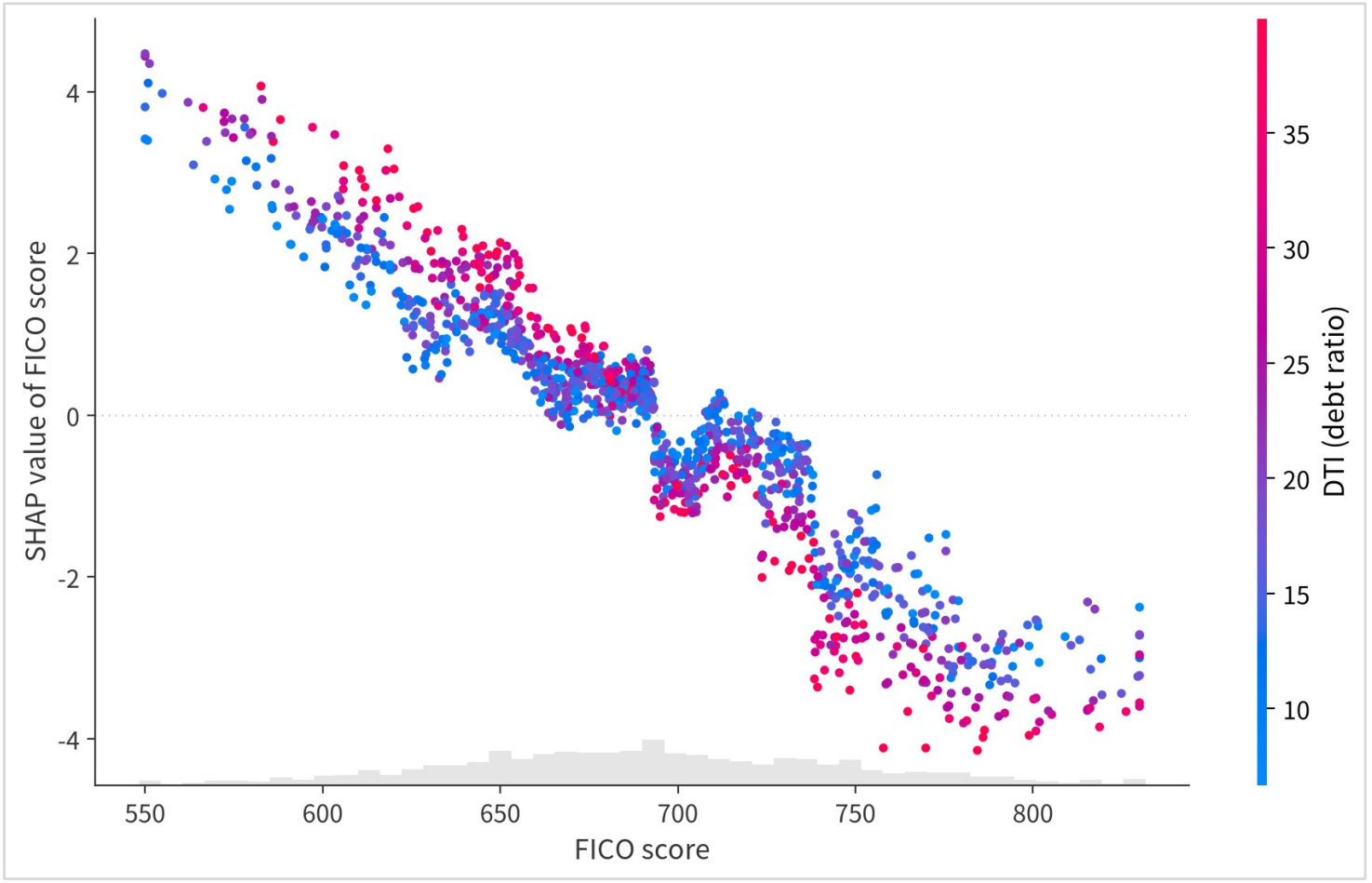


A scatter shows how feature values relate to contributions and interactions

SHAP values of FICO score, colored by DTI. The FICO×DTI interaction built into the data is visible

How to read

- Higher score, lower risk
- Same score, colors split
- At low scores, high DTI sits higher
- Vertical spread is interaction

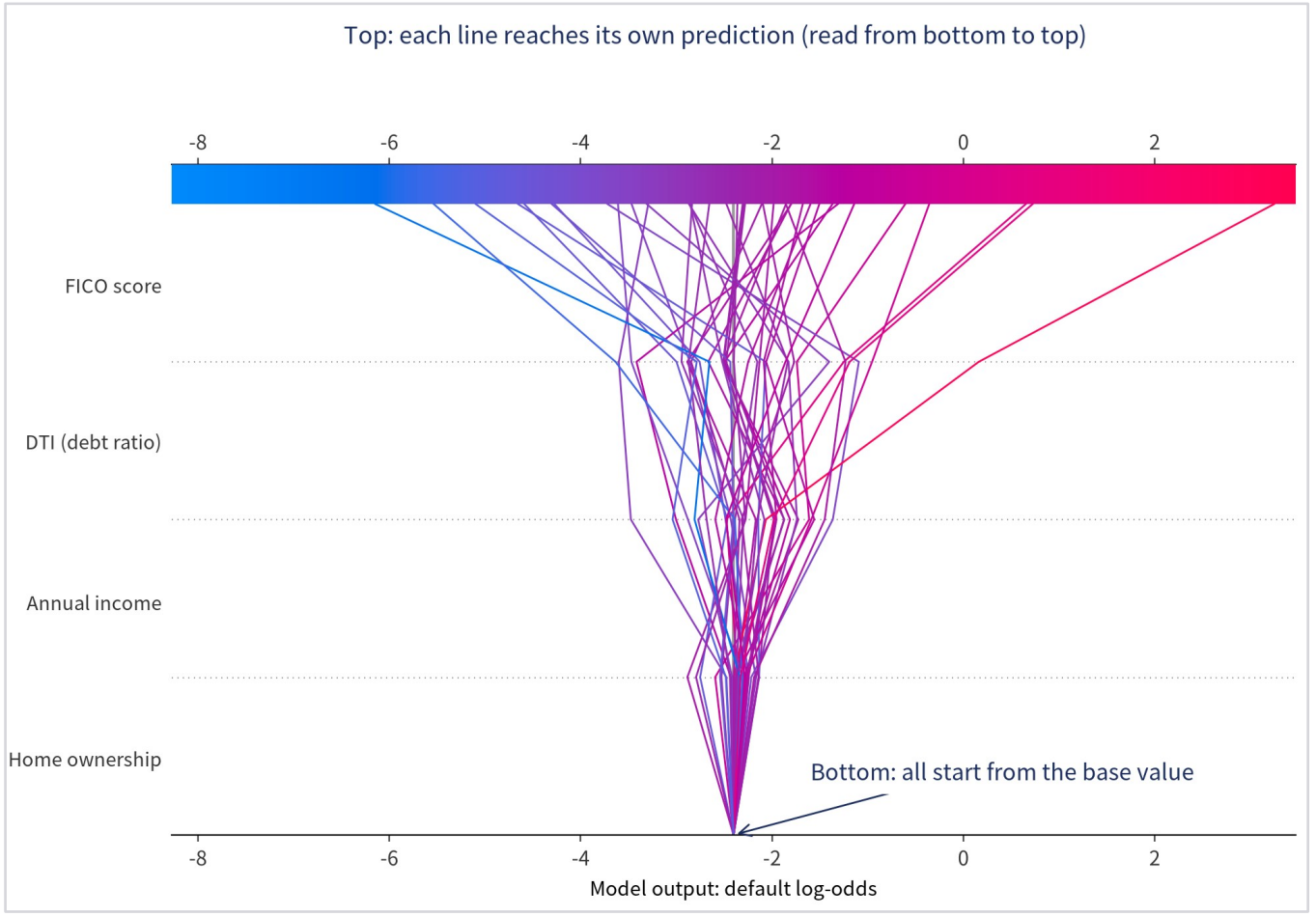


A decision plot starts at the base value at the bottom and accumulates upward

40 customers. All lines meet at the base value at the bottom, then diverge upward to reach their own predictions

How to read

- Read from bottom to top
- Higher features matter more
- Line shift width is contribution
- Top edge is the final prediction



Confirm the reading direction of a decision plot in the official docs

Correcting a mix-up that actually occurred in a July 2025 lecture-prep conversation

The mix-up

- It was said to "read from top to bottom"
- The questioner observed "it converges at the bottom and spreads at the top"
- The questioner was right

What the official docs say

- At the bottom all lines meet at the base value
- At the top each line reaches its own prediction
- Features are stacked upward in order of importance for the displayed samples

Lesson: check a plot's axes and reading direction in the official docs, not from memory

05

PART 5 · Local Explanation

Local surrogates, rules and counterfactual explanations

LIME, Anchors and counterfactuals broaden the forms of explanation

LIME approximates the black box with a simple model around one point

g is an interpretable model; π_x gives large weights to samples close to x (Ribeiro et al., 2016)

$$\xi(x) = \arg\min_{g \in G} \underbrace{L(\hat{f}, g, \pi_x)}_{\text{local fidelity loss}} + \underbrace{\Omega(g)}_{\text{complexity penalty}}$$

1. Perturb

Generate samples around the original data

2. Predict

Predict each sample with the black box

3. Weight

Closer to the original point, larger weight

4. Linear model

The weighted regression coefficients are the explanation

Source: four steps from the Day 4 training slides. Fast, but results can vary with each resampling

Anchors finds sufficient-condition rules that fix the prediction

A is the rule, D(z|A) the distribution of perturbed samples satisfying the rule, τ the target precision

$$\textbf{Precision}(A) = E_{D(z|A)}\left[\mathbf{1}\{\hat{f}(z) = \hat{f}(x)\}\right] \geq \tau$$

Example (earlier lecture)

- "Decline if credit score is below 600 and there are 2+ delinquencies"
- Search for rules at 95% target precision
- Easiest to explain to non-experts

Precision and coverage

- Precision: share of cases with the same prediction when the rule holds
- Coverage: share of samples to which the rule applies
- The two trade off against each other

Counterfactual explanations answer ‘what would have changed the outcome?’

y′ is the desired outcome, d the distance from the original customer (Wachter et al., 2017)

$$x^* = \arg\min_{x'} \max_{\lambda} \underbrace{\lambda\left(\hat{f}(x') - y'\right)^2}_{\text{reach target outcome}} + \underbrace{d(x, x')}_{\text{distance to original customer}}$$

Practical conditions

- Fix immutable features such as age
- Change as few features as possible
- Offer several alternatives, as DiCE does

Why it matters

- It tells customers actions they can actually take
- The CJEU ruling also accepted 'how the result would differ with different inputs' as a means of explanation

The four explanation methods differ in foundation and use

Merges comparison tables from earlier lecture notes. Speed figures that vary by environment were dropped

Criterion	SHAP	LIME	Permutation importance	Counterfactual
Theoretical basis	Shapley axioms	Local surrogate regression	Performance drop	Optimization
Scope	Local and global	Local	Global	Local
Stability	High with exact estimators	Varies by sample	Needs repeated averaging	Multiple solutions
Main use	Reason notices, monitoring	Quick checks	Feature selection	Improvement guidance

06

PART 6 · Deep Learning and LLMs

Interpreting deep learning, time series and LLMs

From gradient-based attribution to circuit tracing in large language models

Integrated gradients integrate the gradient along a path

x′ is the baseline input (e.g. zero vector), α a path variable moving from 0 to 1 (Sundararajan et al., 2017)

$$IG_i(x) = (x_i - x_i') \int_0^1 \frac{\partial \hat{f}\left(x' + \alpha(x - x')\right)}{\partial x_i} d\alpha$$

$$\sum_i IG_i(x) = \hat{f}(x) - \hat{f}(x') \quad \text{(completeness)}$$

Source: deep hedging lecture code

- Measured feature importance with permutation importance and mean absolute gradient

Supplement

- Completeness makes it add up, like SHAP
- DeepSHAP combines DeepLIFT with Shapley values

Attention weights are useful clues but not explanations in themselves

e_t is the score at time t, T the series length

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}$$

Source: advanced lecture notes

- Introduced as showing which time points and news a model attended to in financial time series

Supplement: academic debate

- Jain·Wallace (2019) showed attention weights can diverge from the grounds of prediction
- Wiegreffe·Pinter (2019) countered that they are conditionally useful

LLM interpretation traces internal features and computation paths

A sparse autoencoder (SAE) decomposes internal representation x into interpretable features h

$$h = \mathrm{ReLU}(W_{\mathrm{enc}}x + b), \qquad \mathcal{L} = \underbrace{\|x - W_{\mathrm{dec}}h\|_2^2}_{\text{reconstruction error}} + \underbrace{\lambda\|h\|_1}_{\text{sparsity penalty}}$$

Circuit tracing

- Anthropic open-sourced an attribution graph tool in May 2025
- Usable on open models such as Gemma-2-2b and Llama-3.2-1b

Limits

- Attribution graphs reveal only part of the internal steps
- Still premature as a regulatory tool in finance

Log agent trade rationales in a structured form to trace them

Source: XAI report schema from the April 2026 weekly detailed lecture notes

[Output] Decision rationale report schema · XAIReport

ticker	ticker
final_decision	final trading decision
analyst_basis	4 analyst rationales
debate_summary	bull·bear debate summary
risk_basis	risk assessment basis
fund_manager_note	fund manager approval note
data_sources	data sources used
confidence	decision confidence 0–1

Uses

- Leaves a decision trail for supervisory response and investment committee reports

Caution

- Rationales an LLM writes itself may differ from its actual computation path, so verify them

07

PART 7 · Financial Cases

Financial applications

Credit scoring, fraud detection, asset price prediction, fairness

In fraud detection, SHAP links prediction to improvement

Steps from the FraudDetectionWithSHAP structure in an earlier lecture

1. Predict

A random forest outputs fraud probability

2. Decompose

TreeSHAP yields per-feature contributions

3. Verify

Check base value plus SHAP sum matches the prediction within 0.001

4. Notify

Turn the top 3 factors into customer text and regulatory reports

5. Improve

Analyze SHAP patterns of false positives and negatives

Customer text uses phrases like ‘ratio to usual transaction amount’ instead of technical names (amount_vs_avg, etc.)

A SHAP diagnostic table finds model bloat and bias

analyze_shap_for_model_improvement from an earlier lecture. Thresholds are the source's example values

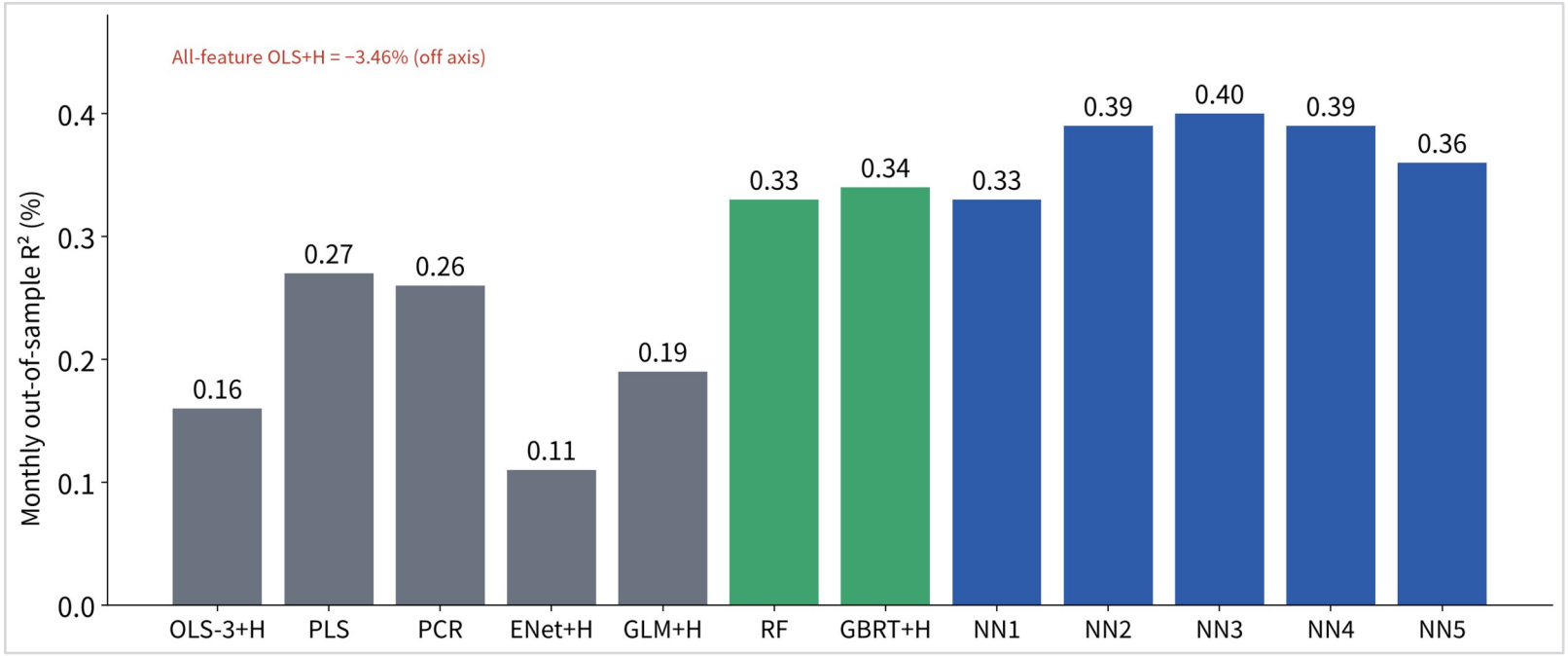
Diagnosis	SHAP signal	Action
Unneeded feature	Mean absolute SHAP value below 0.01	Remove to slim the model and reduce overfitting
One-sided skew	Share of positive SHAP values above 90%	Revisit preprocessing and encoding
Redundant feature	SHAP value correlation between two features above 0.8	Drop one or group them
False positive	Feature with large positive contribution on normal transactions	Adjust threshold and features
False negative	Feature with negative contribution on fraudulent transactions	Strengthen features

Nonlinear models lift out-of-sample R^2 in stock return prediction

Source: Gu, Kelly, Xiu (2020) Table 1, full-sample monthly out-of-sample R^2 (%). Cited in source Big Data Ch. 5

Key findings

- Neural net NN3 0.40%
- Tree models 0.33–0.34%
- OLS on all features −3.46%
- Top signals: momentum, liquidity, volatility



Even without discrimination, failing to explain loses trust

Correcting an earlier lecture's Apple Card case with the regulator's findings

Apple Card (2019~2021)

- In 2019 a credit-limit gap between spouses sparked a gender bias controversy
- In 2021 NYDFS reviewed data on about 400,000 people and found no evidence of violation
- It noted, however, that lack of transparency hurt trust

Amazon hiring AI (2018)

- A model trained on past résumés penalized female applicants and was scrapped
- Proxies for protected attributes, such as ZIP codes, create the same risk

If a proxy shows up among top SHAP reasons, measure fairness metrics separately on actual outcomes

08

PART 8 · Limits and Regulation

Limits, pitfalls, and the 2026 regulatory landscape

What to check before trusting an explanation, and what to comply with

In practice, explanation methods often reach conflicting conclusions

Krishna et al. (2022), interviews with 25 data scientists. The ‘disagreement problem’

88%

Use several explanation methods on the same prediction

84%

Frequently encounter disagreement between methods

86%

Resolve it ad hoc or don't know how (user study)

Forms of disagreement

- Different top features, or different order and sign

Response

- Choose the method for the purpose in advance and document it

Perturbation-based explanations can be deliberately fooled

Slack et al. (2020), 'Fooling LIME and SHAP', AIES 2020

**A biased classifier can be wrapped so that, only when explanation perturbations arrive,
it answers like a harmless model.**

Mechanism (scaffolding)

- Perturbed samples lie off the real data distribution
- Detect off-distribution inputs and answer with a different model only then
- Both LIME and SHAP were fooled on real data such as COMPAS

Implications

- Perturbation-based explanations alone cannot detect bias
- Measure fairness directly on outcomes over real data
- For external models, obtain outcome distributions along with explanations

Five pitfalls are easy to fall into when using SHAP

Cautions scattered across the source lectures collected in one table

Pitfall	Description	Response
Correlated features	Contributions split or swap	Aggregate as groups and state the method
Causal illusion	Model association, not real-world causation	Assess intervention effects with causal analysis
Base value dependence	Values change with background data	Document the background sample
Output scale	Log-odds and probabilities differ in magnitude	State units on every chart
Approximation error	Kernel SHAP varies with few samples	Check additivity and repeat

Overseas rules require explaining the grounds of individual decisions

Key EU and US bases, as of October 2026

Jurisdiction · basis	Explanation requirement	Current status
EU AI Act	Credit scoring is high-risk (Annex III). Right to explanation for affected persons (Art. 86)	High-risk obligations postponed to 2 Dec 2027 (Digital Omnibus)
GDPR Art. 15 + CJEU C-203/22	Explain the procedures and principles actually applied in an understandable way	Ruling of 27 Feb 2025
US ECOA · Regulation B	Notice of specific principal reasons for credit denial	Two CFPB interpretive letters withdrawn May 2025; rule remains

In Korea, too, the right to explanation of automated decisions is established

AI Basic Act effective 22 Jan 2026; Financial AI Guidelines effective 22 Jun 2026

Basis	Key content
AI Basic Act	Automated loan screening without final human judgment is high-impact AI. Fines up to KRW 30 million; 1+ year grace period
PIPA Art. 37-2	Data subjects' right to request explanation of, and refuse, fully automated decisions
Credit Information Act Art. 36-2	Right to request explanation of and object to automated assessment results
Revised Financial AI Guidelines	Seven principles: governance, legality, assistive role, reliability, financial stability, good faith, security

CJEU: an understandable explanation, not the algorithm

Practical implications of C-203/22 Dun & Bradstreet Austria, ruling of 27 Feb 2025

Don't

- Hand over complex formulas or the whole algorithm
- List every step of the entire process
- Refuse wholesale citing trade secrets

Do

- State which personal data were used and how
- Show how the result would differ with different inputs
- Trade secret disputes are weighed by the supervisor or court

Top SHAP reasons combined with counterfactuals come closest to the explanation the ruling describes

Explanation documents need seven items, from purpose to limits

A checklist covering both model risk management and supervisory response

What is explained

- Purpose of use and scope of automated decisions
- Main factors and direction of effect
- Explanation methodology and rationale for the choice
- Background data and output units

How far it can be trusted

- Known limits of the explanation
- Example explanations for representative cases
- Records of re-review at each retraining

Roll out the explanation system in three phases

Phase 1–3 roadmap from an earlier lecture

Phase 1 · Top factors

Basic

Notify the top 3 factors by absolute SHAP.
Automate the additivity check.

Customer inquiry response

Phase 2 · What-if analysis

What-if

Simulate "What if income rises?"
Fix immutable features.

Counterfactual guidance

Phase 3 · Natural-language explanation

LLM text generation

Turn SHAP results into sentences.
The system fills in numbers; the LLM writes only the sentences.

Hallucination prevention

Discussion Questions

Organize your judgment with the principles covered in class

Q1	Two highly correlated income variables split the #1 and #2 rejection reasons. What would you tell the customer, and how?	**Correlated features**
Q2	‘Residential area’ appears among a loan model's top SHAP reasons. Is that itself evidence of discrimination? What else would you check?	**Fairness**
Q3	SHAP and LIME give different #1 reasons for the same customer. What would you use for the notice and for internal reports?	**Disagreement**
Q4	Can a trading rationale report written by an LLM agent itself serve as supervisory material?	**Self-explanation**

Explanation is decomposition, approximation, and obligation

Today's lecture in three sentences

1

Decomposition

Thanks to the Shapley axioms, SHAP splits a prediction exactly into a base value plus contributions.

2

Approximation

Use it knowing the background distribution, correlation, disagreement between methods, and manipulability.

3

Obligation

Regulation requires not algorithm disclosure but explanations customers can understand and act on.

Next: in the workbook, compute three-feature Shapley values and permutation importance by hand

References and Sources

Sources are the past lecture materials merged into this lecture

- Sources: SHAP Analysis for Tree Models (2025.7), financial process automation lecture (2025.8), financial AI guidelines lecture notes (2025.10), Big Data Ch. 5 machine learning translation (2025.8), Day 4 training slides (2025.6), advanced lecture notes (2025.8), 8-week deep hedging course (2025.8), weekly detailed lecture notes (2026.4)
- Shapley (1953); Lundberg·Lee (2017) NeurIPS; Lundberg et al. (2020) Nature Machine Intelligence; Ribeiro et al. (2016, 2018); Wachter et al. (2017); Sundararajan et al. (2017)
- Apley·Zhu (2020); Chen et al. (2020); Janzing et al. (2020); Muschalik et al. (2024) shapiq; Gu·Kelly·Xiu (2020) Review of Financial Studies
- Krishna et al. (2022) The Disagreement Problem; Slack et al. (2020) Fooling LIME and SHAP; Jain·Wallace (2019); Wiegreffe·Pinter (2019); Anthropic (2025) circuit tracing
- shap official docs (decision plot), PyPI shap 0.54.1; Regulation (EU) 2026/1744; CJEU C-203/22; NYDFS Apple Card report (2021.3); Aju Business Daily · New Daily (2026) coverage of the AI Basic Act and Financial AI Guidelines